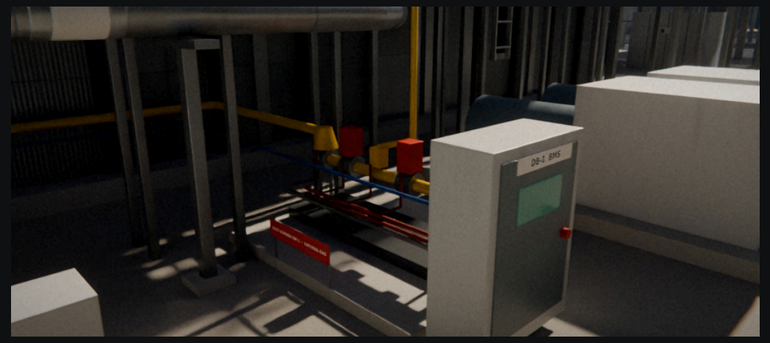
BOILERS: GREENFIELD & BROWNFIELD

One cable partner from the new HRSG to the package boiler to the outage that replaces a module: the same products, staged and pulled the same way.



01 · GREENFIELD

HEAT-RECOVERY BOILER

Behind every gas turbine: three-pressure HRSG with SCR and a duct burner. Feed pumps at 13.8 kV, drum-level and thermocouple runs, burner management in a Class I Div 2 skid.

705
CABLES PER HRSG TRAIN

156,000
FT OF CABLE PER TRAIN



02 · GREENFIELD OR RETROFIT

GAS-FIRED PACKAGE BOILER

The plant's auxiliary boiler, the same package sold to industry: burner and BMS, gas train with double block and bleed, FD fan on a VFD, feedwater pumps, its own MCC e-house.

83
CABLES PER PACKAGE BOILER

42,000
FT OF CABLE



03 · BROWNFIELD

MODULE REPLACEMENT OUTAGE

An HP evaporator module lifted out through the opened roof. The cable work rides the same outage window, so it has to be staged, pre-terminated and pulled fast.

Outage
CRITICAL PATH

SIMpull
+ PRE-TERMINATED HARNESSES

CABLE BY BOILER TYPE

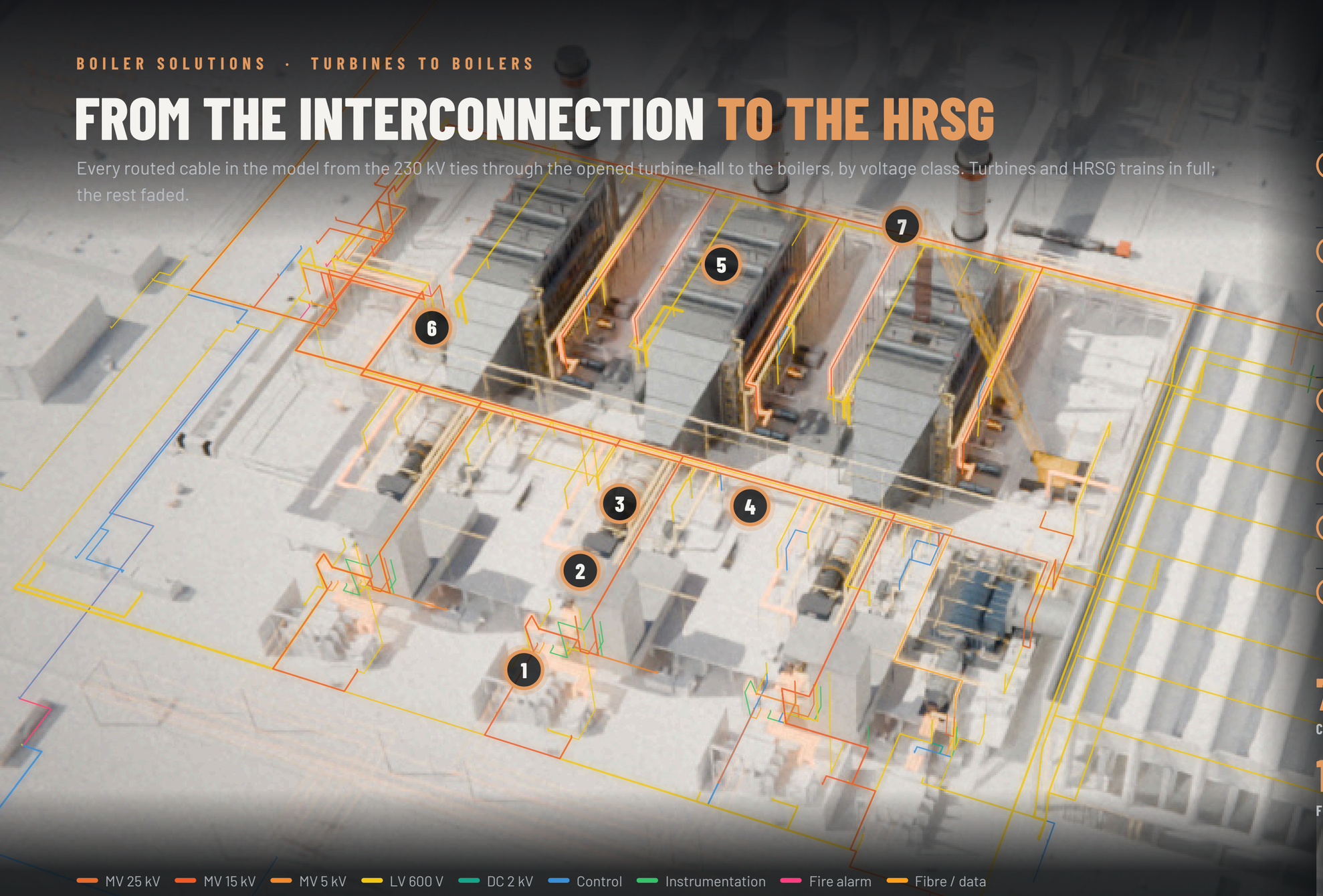
	MV-105 15 KV FEED PUMPS	TC-ER 600 V POWER & VFD	ARMOR-X MC-HL GAS TRAINS (C1D2)	CONTROL TC-ER	INSTRUMENTATION PLTC · TC EXT.	SIMPULL · REEL ROLLERS	PRE-TERMINATED HARNESSES
HRSG + DUCT BURNER	●	●	●	●	●	●	●
PACKAGE BOILER		●	●	●	●	●	●
REPLACEMENT OUTAGE	○	●	●	●	●	●	●

● in scope ○ when the outage includes feed-pump or MV work

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED · QUANTITIES FROM THE MODEL'S CABLE SCHEDULE

FROM THE INTERCONNECTION TO THE HRSG

Every routed cable in the model from the 230 kV ties through the opened turbine hall to the boilers, by voltage class. Turbines and HRSG trains in full; the rest faded.



1. **GSUS AND 230 KV TIES**
   step-up transformers, the lines to the switchyard
2. **GENERATORS AND 21 KV BUS**
   IPB, unit aux transformers, excitation
3. **GAS TURBINES**
   instrument, control and aux power on every unit
4. **HALL MV AND LV TRAYS**
   13.8 kV feeders, MCC power, control
5. **HRSG TRAINS**
   feed pumps, duct burners, drums, SCR, CEMS
6. **HRSG E-HOUSES**
   MCCs and DCS I/O per unit
7. **MAIN CABLE RACK**
   MV, LV, control and fibre to the units

705
CABLES PER HRSG TRAIN

156,000
FT OF CABLE PER TRAIN

MV 25 kV · MV 15 kV · MV 5 kV · LV 600 V · DC 2 kV · Control · Instrumentation · Fire alarm · Fibre / data

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED · QUANTITIES FROM THE MODEL'S CABLE SCHEDULE



HRSG 3: an HP evaporator module lifted out through the opened roof by a crawler crane; the new module waits on a lowboy on the ring road.

BOILER SOLUTIONS · REPLACEMENT

THE OUTAGE CLOCK IS THE COST

On a boiler or HRSG replacement every hour of the outage costs. Cable rides the same window, so it moves off the critical path.

- **STAGE BEFORE THE TRIP** cut-to-length reels on SIMpull Cradles and reel rollers, labelled per pull, at the unit before the outage starts
- **PRE-TERMINATE** factory-made harnesses with QR tags land on the marshalling boxes: test, not terminate, on the critical path
- **PULL CLEAN** SIMpull reels and a sheave at the HRSG wall: one crew, no reel jacks, tension recorded
- **CUT BACK AND RECYCLE** old cable segregated into recycling bins as the module comes out
- **CONVERSIONS** new burner management and gas trains in Class I Div 2 run in ARMOR-X MC-HL



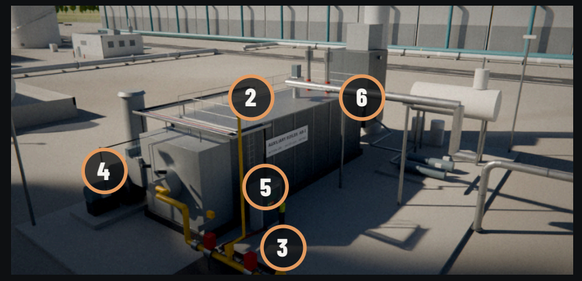
Cable work at the HRSG e-house



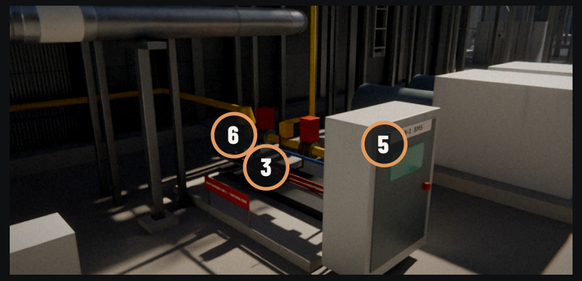
Old module on the hook

stephan.hardt@southwire.com

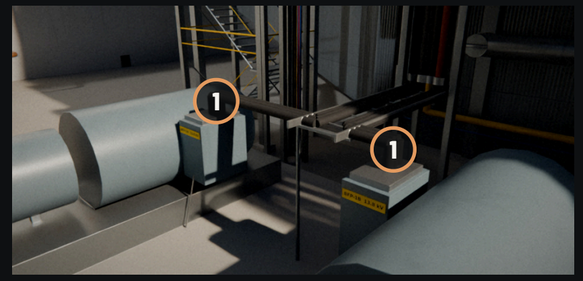
THE CABLE AT EVERY SPOT



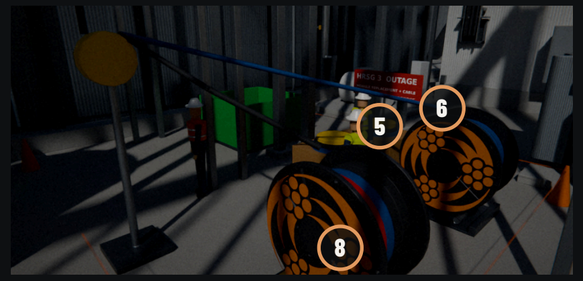
Package boiler AB-1



HRSG duct-burner skid



HRSG feed pumps, 13.8 kV



Outage cable work, HRSG 3

	PRODUCT	WHERE IT RUNS	SPEC	SOUTHWIRE STOCK # · SIZE	SPEC SHEET
1	**MV-105 15 KV** 1/C Cu, 220 mil NL-EPR 133 %, tape shield, SIMpull PVC	Feed-pump motors, 13.8 kV	46302	95604599 · 500 kcmil	southwire.com ›
2	**TYPE TC-ER POWER** 3/C Cu XHHW-2 + ground, PVC, 600 V	Fans, pumps, MCC feeders	45252	95305999 · 12 AWG 3/C	southwire.com ›
3	**ARMOR-X MC-HL** Cu XLPE XHHW-2, welded armor, PVC, 600 V, Class I Div 2	Gas trains, burner skids	45352	89052799 · 8 AWG 4/C	southwire.com ›
4	**ARMOR-X MC-HL VFD** 3/C Cu XLPE XHHW-2 + ground, VFD rated	FD fan on its drive	45352	89051899 · 3/0 3/C	southwire.com ›
5	**CONTROL TC-ER** Cu XLPE XHHW-2, 600 V	BMS panels, valves, MOVs	45053	89067701 · 14 AWG 5/C	southwire.com ›
6	**INSTRUMENTATION PLTC / ITC** 300 V, shielded pairs + overall shield (SPOS)	Flame scanners, drum level, transmitters	43303	per pair count	southwire.com ›
7	**THERMOCOUPLE EXTENSION** PVC, PLTC multi-pair (KX on request), 105 °C	HRSG gas-path and tube temperatures	42345	per construction	cabletechsupport.southwire.com ›
8	**SIMPULL CRADLE** for 51 / 61 in SIMpull Reels, one-person payoff	Outage staging and pulls	SPR-CRDL	67606001	southwire.com ›

Stock numbers are one catalogue size per product, from Southwire's product listings; every size and construction is on the same SPEC. Confirm current stock with Southwire customer service before quoting.

stephan.hardt@southwire.com